

# Jishnu Nambissan Thekkeppattu (he/him)

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## CONTACT INFORMATION

CSIRO Space and Astronomy,  
26 Pembroke Road,  
Marsfield, NSW 2122, Australia.  
[jishnunambissan.thekkeppattu@csiro.au](mailto:jishnunambissan.thekkeppattu@csiro.au)  
<https://jnthek.github.io/>

## RESEARCH INTERESTS

Radio astronomy, radio interferometry, instrumentation and techniques. Phased arrays, calibration and array signal processing, wide-field polarimetry. High speed DSP, reconfigurable computing, FPGAs. Experimental and observational 21-cm cosmology, precision radiometry, data analysis for 21-cm experiments, radio-sky modelling at low-frequencies. All-sky spherical harmonic transit interferometry (*m*-mode imaging). Numerical linear algebra, ill-posed inverse problems, iterative methods. Ionospheric scintillation.

## PROFESSIONAL EXPERIENCE

|                     |                                                                        |
|---------------------|------------------------------------------------------------------------|
| Aug 2025 - Present  | CERC Fellow, CSIRO Space and Astronomy, Australia                      |
| Aug 2022 - Jul 2025 | Research Associate, Curtin University, Australia                       |
| Nov 2016 - Jul 2022 | Research Scientist/Research Associate, Raman Research Institute, India |
| Jan 2016 - Oct 2016 | Assistant System Engineer, Tata Consultancy Services, India            |

## EDUCATION

### Curtin University, Australia

Doctor of Philosophy – Physics (2019–2022)

- Thesis title : Towards detection of redshifted 21-cm signal from Cosmic Dawn and Epoch of Reionization.
- Thesis supervisors : Dr. Cathryn M. Trott, Dr. Benjamin McKinley (Curtin Institute of Radio Astronomy, Bentley, WA)

### Kannur University, India

Bachelor of Technology in Electronics and Communication Engineering (2011–2015)

- Aggregate percentage : 81.2%
- Undergraduate project : A VHF board for Software Defined Radio (SDR) applications.
- Project advisor: Dr. V. Vinodkumar (Govt. College of Engineering, Kannur)

## PUBLICATIONS ORCID : [0000-0002-3079-4046](https://orcid.org/0000-0002-3079-4046)

### First authored, peer reviewed, journal papers

**Jishnu N. Thekkeppattu**, Randall Wayth, and Marcin Sokolowski, *Calibration of an SKA-Low prototype station using holographic techniques*, 2024, Radio Science <https://doi.org/10.1029/2023RS007847>.

**Jishnu N. Thekkeppattu**, Cathryn M. Trott, and Benjamin McKinley, *Singular spectrum analysis of time series data from low frequency radiometers, with an application to SITARA data*, 2023, Monthly Notices of the Royal Astronomical Society, <https://doi.org/10.1093/mnras/stad522>.

**Jishnu N. Thekkeppattu**, Benjamin McKinley, Cathryn M. Trott, Jake Jones, Daniel C. X. Ung, *System design and calibration of SITARA - a global 21-cm short spacing interferometer prototype*, 2022, Publications of the Astronomical Society of Australia, <https://doi.org/10.1017/pasa.2022.13>

**Jishnu Nambissan, T.**; Ravi Subrahmanyam; R. Somashekar; N. Udaya Shankar; Saurabh Singh; A. Raghunathan; B. S. Girish; K. S. Srivani; Mayuri Sathyanarayana Rao *SARAS 3 CD/EoR radiometer: design and performance of the receiver*, 2021, Experimental Astronomy, <https://doi.org/10.1007/s10686-020-09697-2>

### Co-authored, peer reviewed, journal papers

Macario, Giulia; Pupillo, Giuseppe; Bernardi, Gianni; Di Ninni, Paola; Comoretto, Giovanni; Bolli, Pietro; Mattana, Andrea; Monari, Jader; Perini, Federico; Schiaffino, Marco; Sokolowski, Marcin; Wayth, Randall; **Thekkeppattu, Jishnu N.**, *Intrinsic cross-polarization ratio maps from all-sky observations with the SKA-Low prototype station Aperture Array Verification System 2*, Journal of Astronomical Telescopes, Instruments, and Systems, Volume 11, id. 028008 (2025) <https://doi.org/10.1117/1.JATIS.11.2.028008>

Saurabh Singh, **Jishnu Nambissan T.**, Ravi Subrahmanyam, N. Udaya Shankar, B. S. Girish, A. Raghunathan, R. Somashekar, K. S. Srivani, Mayuri Sathyanarayana Rao, *On the detection of a cosmic dawn signal in the radio background*, Nature Astronomy, 6, 607-617 (2022) <https://doi.org/10.1038/s41550-022-01610-5>

Girish, B. S.; Srivani, K. S.; Subrahmanyam, Ravi; Udaya Shankar, N.; Singh, Saurabh; **Jishnu Nambissan, T.**; Sathyanarayana Rao, Mayuri; Somashekar, R.; Raghunathan, A. *SARAS CD/EoR Radiometer: Design and Performance of the Digital Correlation Spectrometer*, 2020, Journal of Astronomical Instrumentation, Volume 9, Issue 2. <https://doi.org/10.1142/S2251171720500063>

B McKinley, C M Trott, M Sokolowski, R B Wayth, A Sutinjo, N Patra, **J. Nambissan T.**, D C X Ung, The All-Sky SignAl Short-Spacing INterferometer (ASSASSIN) - I. Global-sky measurements with the Engineering Development Array-2, Monthly Notices of the Royal Astronomical Society, Volume 499, Issue 1, November 2020, Pages 52-67, <https://doi.org/10.1093/mnras/staa2804>

Raghunathan, A.; Subrahmanyam, R; Shankar, N. U.; Singh, S; **Nambissan, J.**; Kavitha, K.; Mahesh, N.; Somashekar, R.; Sindhu, G; Girish, B. S.; Srivani, K. S.; Rao, M. S., *A Floating Octave Bandwidth Cone-Disk Antenna for Detection of Cosmic Dawn*, in IEEE Transactions on Antennas and Propagation, vol. 69, no. 10, pp. 6209-6217, Oct. 2021, <https://doi.org/10.1109/TAP.2021.3069563>

Adrian T. Sutinjo, Benjamin McKinley, Leonid Belostotski, Daniel C. X. Ung, and **Jishnu Nambissan T.**, *Design Equations for Closely Spaced Two-Element Interferometer for Radio Cosmology*, URSI Radio Science Letters, Vol. 2, 2020, <https://doi.org/10.46620/20-0017>

### Conference Proceedings

**Thekkeppattu, J. N.**, Cook, J. H., Wayth, R. B., Carozzi, T. D., Sokolowski, M., Kriele, M. A., and Vijayaraghavan, M. (2025). *All-sky m-mode map making at 159 MHz with EDA2 and SE607HBA LOFAR station*. In Proceedings of the 7th URSI Asia-Pacific Radio Science Conference - AP-RASC 2025. 7th URSI Asia-Pacific RadioScience Conference. URSI - International Union of Radio Science. <https://doi.org/10.46620/ursiaprasc25/wqun6574>

Cook, J., **Thekkeppattu, J.**, Wayth, R., and Kriele, M. (2025). *All Sky maps from m-mode Spherical Harmonic Transit Interferometry with the MWA and the EDA2*. In

Proceedings of the 7th URSI Asia-Pacific Radio Science Conference - AP-RASC 2025. 7th URSI Asia-Pacific Radio Science Conference. URSI - International Union of Radio Science.

<https://doi.org/10.46620/ursiaprasc25/pomb8661>

**Thekkeppattu, J. N.**, Sokolowski, M., and Wayth, R. B. (2025). *Broadband self-holography of EDA2 - an SKA-Low prototype station*. In Proceedings of the 7th URSI Asia-Pacific RadioScience Conference - AP-RASC 2025. 7th URSI Asia-Pacific Radio Science Conference. URSI International Union of Radio Science.

<https://doi.org/10.46620/ursiaprasc25/hech4944>

Raghunathan, Agaram; Girish, B.S.; Somashekar, R.; Srivani, K.S.; Singh, Saurabh; Subrahmanyam, Ravi; Shankar, N. Udaya; Rao, Mayuri Sathyanarayana; **Nambissan T., Jishnu**, *Wide band antenna with Ultra-smooth spectral characteristics*, 2020 International Applied Computational Electromagnetics Society Symposium (ACES), 2020, pp. 1-2, <https://doi.org/10.23919/ACES49320.2020.9196075>

**Nambissan, T. J.**, Nikhil, T., Vinodkumar, V., *A VHF Radio for Software Defined Radio Applications*, 2016, Procedia Technology, 24, 820, International Conference on Emerging Trends in Engineering, Science and Technology (ICETEST - 2015)

<https://doi.org/10.1016/j.protcy.2016.05.109>

#### Non peer reviewed articles / pre-prints

**Jishnu N. Thekkeppattu**, *First results from a low cost software defined radio system monitoring VHF equatorial ionospheric scintillations in the Indian sector*, 2024, [ESS open archive](#).

RESEARCH  
EXPERIENCE

- 2025-Present    Development of an ultrafast radio camera using FPGAs  
Supervisor(s): Dr. Nithyanandan Thyagarajan, Dr. David Humphrey  
CSIRO Space and Astronomy, Marsfield, NSW.
- 2022-2025       SKA-Low bridging and Early Production Array activities &  
All-sky mapping with spherical harmonic transit interferometry  
Supervisor(s): Dr. Randall Wayth, Dr. David Davidson  
Curtin Institute of Radio Astronomy, Bentley, WA.
- 2019-2022       Towards detection of redshifted 21-cm signal from Cosmic Dawn  
and Epoch of Reionization.  
Supervisor(s): Dr. Cathryn Trott, Dr. Benjamin McKinley,  
Curtin Institute of Radio Astronomy, Bentley, WA.
- 2019-2021       Principal investigator of PRATUSH: Probing ReionizATIOn of the  
Universe using Signal from Hydrogen.  
(Funded for pre-project studies by the ISRO)  
Co-PIs : Dr. Saurabh Singh, Dr. Mayuri S. Rao  
Raman Research Institute, Bangalore, India.
- 2016-2021       On the detection of global 21-cm signal from Cosmic Dawn and  
Epoch of Reionization.  
Advisors: Dr. Ravi Subrahmanyam, Dr. N. Udaya Shankar,  
Raman Research Institute, Bangalore, India.
- 2019    June-    Comparing LOFAR Epoch of Reionization target fields using 21-cm  
August    power spectra.  
Advisor : Dr. André Offringa,  
ASTRON, Dwingeloo, The Netherlands.
- 2014-2015       Reconfigurable radios using Field Programmable Gate Arrays.  
Advisor: Dr. V. Vinodkumar,  
Govt. College of Engineering, Kannur, India
- 2014-2015       A VHF board for Software Defined Radio (SDR) applications.  
Advisor: Dr. V. Vinodkumar,  
Govt. College of Engineering, Kannur, India
- 2013-2014       Multiplierless filter realisation using Canonic Signed Digits.  
Advisor: Dr. Manju Manuel,  
Govt. College of Engineering, Kannur, India

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|-----------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SELECTED TALKS              | 2026 Aug               | <i>Monitoring equatorial ionospheric scintillation using low-cost software defined radios - some results from the Indian sector</i> - URSI-GASS-2026, held at Krakow, Poland (YSA talk)                           |
|                             | 2026 Aug               | <i>Ultrafast real-time imaging with large-N aperture arrays using FPGA accelerators</i> - URSI-GASS-2026, held at Krakow, Poland                                                                                  |
|                             | 2025 Dec               | <i>Ultra-fast real-time imaging with large aperture arrays</i> - SALF-XI, held at Orleans, France (remote talk)                                                                                                   |
|                             | 2025 Aug               | <i>All-sky m-mode map making at 159 MHz with EDA2 and SE607HBA LOFAR station</i> - URSI-AP-RASC 2025, held at Sydney, Australia                                                                                   |
|                             | 2025 Aug               | <i>Broadband self-holography of EDA2 - an SKA-Low prototype station</i> - URSI-AP-RASC 2025, held at Sydney, Australia                                                                                            |
|                             | 2025 Jun               | <i>All-sky map making with spherical harmonics in the SKA-Low era</i> - SKAO General Science Meeting, A new era in astrophysics: Preparing for early science with the SKAO, held at Goerlitz, Germany             |
|                             | 2020 Jan               | <i>Updates from the SARAS experiment</i> - International School And Conference On Observing The First Billion Years Of The Universe Using The Next Generation Telescopes, held at IIT Indore, India (remote talk) |
| HONORS AND AWARDS           | 2026                   | Young Scientist Award, URSI-GASS-2026                                                                                                                                                                             |
|                             | 2022                   | Joint winner of the ICRAR Ken and Julie Michael Prize                                                                                                                                                             |
|                             | 2014                   | Best student paper award in embedded systems and VLSI category at National Conference on Systems Energy and Environment - 14 held at Govt College of Engineering, Kannur                                          |
|                             | 2010                   | Indian Space Research Organisation (ISRO) sponsored student delegate, 97 <sup>th</sup> Indian Science Congress held at Kerala University.                                                                         |
| TEACHING                    | 2025                   | COMT4003 - Mobile Radio Communications; Unit Co-ordinator & Lecturer (Curtin University)                                                                                                                          |
|                             | 2024                   | COMT6005 - Satellite Communications; Lecturer (Curtin University)                                                                                                                                                 |
|                             | 2024                   | COMT4003 - Mobile Radio Communications; Lecturer (Curtin University)                                                                                                                                              |
|                             | 2023                   | COMT6005 - Satellite Communications; Lecturer (Curtin University)                                                                                                                                                 |
|                             | 2020                   | ELEN3002/6017 - Engineering Electromagnetic Labs (Curtin University)                                                                                                                                              |
| MEMBERSHIPS                 | 2023-Present           | <a href="#">Junior member</a> - International Astronomical Union (IAU)                                                                                                                                            |
|                             | 2024-2025              | Member - Institute of Electrical and Electronics Engineers (IEEE)                                                                                                                                                 |
|                             | 2013-2014              | Student member - IEEE                                                                                                                                                                                             |
| SOFTWARE TOOLS AND PACKAGES | Programming Languages: | python, C, C++, <b>bash</b> scripting                                                                                                                                                                             |
|                             | HDL:                   | VHDL                                                                                                                                                                                                              |
|                             | Software packages:     | AMD Vivado, MATLAB, <b>miriad</b>                                                                                                                                                                                 |
|                             | Operating systems:     | Linux, Mac OS                                                                                                                                                                                                     |
|                             | Misc:                  | x86 and 8051 assembly, embedded C, arduino, Visual C#<br>Verilog, EM simulators (NEC2, CST) and HPC systems                                                                                                       |

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|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBSERVING AND<br>FIELD EXPERIENCE        | Deployment, operation and maintenance of a prototype two-element broadband interferometer SITARA at Inyarrimanha Ilgari Bundara (the Murchison Radio-astronomy Observatory) (2021-2022).                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                          |
|                                          | Measurement of global sky brightness temperature at low frequencies deploying SARAS at remote sites at Timbaktu Collective, India (2017-18), Indian Astronomical Observatory, Ladakh, India (2017-18) and Mavathur lake, India (2019).                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                          |
|                                          | Electromagnetic tests and measurements of performance and characteristics of the SARAS antennas and receivers at the Gauribidanur Field Station, India (2017-19) and Mavathur lake, India (2019).                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                          |
| HARDWARE AND<br>LABORATORY<br>EXPERIENCE | Expertise in electronics and RF/microwave systems; using vector network analysers (Keysight FieldFox, PNA-X etc.) and their calibration, spectrum analysers, system level RF design, development and testing. Hands-on experience in handling fibre-optics, optical power measurements and RFoF systems. Experienced in performing measurements in an electromagnetic compatibility (EMC) chamber. Expertise in basic laboratory/workshop tasks such as soldering and miscellaneous mechanical works. FPGA systems (Zynq, Alveo v80). |                                                                                                                                                                                          |
| OUTREACH                                 | 2025–Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CSIRO S&A Marsfield colloquia organiser.                                                                                                                                                 |
|                                          | 2020–2025                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | DIY radio telescope programme. In collaboration with the Curtin ASTRO-3D node, built a low-cost hobby radio telescope operating at 1420 MHz for outreach purposes.                       |
|                                          | 2015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Institute of Electrical and Electronics Engineers - Teachers In Service programme (IEEE-TISP) volunteer. Conducted science and engineering outreach programmes for high school students. |
| VOLUNTEERING<br>AND HOBBIES              | 2021-Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Australian Red Cross Lifeblood blood, platelets & plasma donor                                                                                                                           |
|                                          | 2014-Present                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Amateur radio operator, holding Indian callsign VU3VWB and Australian callsign VK6JN                                                                                                     |